Analysis of Munich Neighbourhoods for New Immigrants Using Machine Learning

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1. Introduction

1.1. Background

When I began this project, I had just landed in Munich, Germany in January 2026 as a brand new immigrant. One of the first statistics I came across stopped me in my tracks: Germany recorded over 262,000 net immigrants in 2024, with Munich consistently ranking as one of the top destination cities for newcomers. [1] A 2025 Mercer Quality of Living Survey ranked Munich as the 3rd best city in the world to live in [2] so it is no surprise that thousands of people from across the globe choose to settle here every year.

Munich has 25 Stadtbezirke (city districts), and they are not all equal. Crime rates vary. Unemployment varies. Rent varies dramatically. As a new immigrant and data analyst, I decided to stop guessing and start analysing.

This project was inspired by Jessica Ayodele's analysis of Toronto Neighbourhoods. [3]

1.2. Problem Statement

Munich's 25 Stadtbezirke span the entire city from the historic Altstadt at its centre to the quieter outer districts on its edges. As a new immigrant, the vital question to answer is:

"What district do I settle in?"

The aim of this project is to group Munich's 25 districts in order of desirability for new immigrants using Machine Learning and Data Visualisation techniques.

1.3. Basis

There are several factors to consider when settling down in any location. For this project, I performed my analysis using the following criteria:

- Primary Benchmarks: Unemployment rate and Crime rate per district
- Secondary Benchmark: Average monthly rent for a 1-bedroom apartment per district (EUR)

2. Data

2.1. Data Description

For this project, all datasets were obtained from official German and Munich open data portals:

- Crime Rate Data: Crimes per 1,000 residents per district from the Kriminalitatsatlas Munchen (Munich Crime Atlas) 2023. [4]
- Unemployment Rate Data: Percentage of unemployed residents per district from the Statistisches Amt Munchen, Social Atlas 2023. [5]
- Housing Rental Prices: Average monthly rent for a 1-bedroom apartment per district from Immoscout24 rental market report, 2024. [6]
- District Geographic Coordinates: Latitude and longitude point coordinates for all 25 Munich Stadtbezirke via OpenStreetMap and GeoPandas. [7]

2.2. Data Cleaning

The Python modules used for Data Wrangling and Manipulation were Pandas and GeoPandas. GeoPandas was used for spatial coordinates while Pandas was used to handle all other district datasets. All three datasets were merged on the district_id column to produce one final dataframe. See Fig 2.0.

	district_id	district_name	unemployment_rate	crime_rate	avg_rent_eur
0	1	Altstadt-Lehel	4.2	98.2	2100
1	2	Maxvorstadt	3.8	42.1	1950
2	3	Schwabing-West	3.5	35.4	1850
3	4	Schwabing-Freimann	3.2	28.6	1750
4	5	Au-Haidhausen	4.8	52.3	1800
5	6	Sendling	5.2	38.7	1550
6	7	Sendling-Westpark	4.6	31.2	1500
7	8	Schwanthalerhöhe	5.8	45.6	1650
8	9	Neuhausen-Nymphenburg	3.4	29.8	1900
9	10	Moosach	5.6	41.3	1450

Fig 2.0: Pandas Dataframe showing Munich districts and primary benchmarks

3. Methodology

In this section, all Python libraries and techniques used in this project are discussed. The libraries used in the Google Colab notebook are listed below:

- Pandas - For storing and manipulating structured data. Pandas functionality is built on NumPy
- NumPy - For multi-dimensional array and mathematical operations
- GeoPandas - For storing spatial data coordinates using gpd.points_from_xy()
- Scikit-learn - For Machine Learning tasks: StandardScaler for feature scaling and KMeans for clustering
- Plotly Express - For all interactive visualisations saved as HTML files and hosted on GitHub Pages
- Matplotlib - For early exploratory bar charts during the data analysis phase

The main steps for this project can be summarised in the flowchart below:

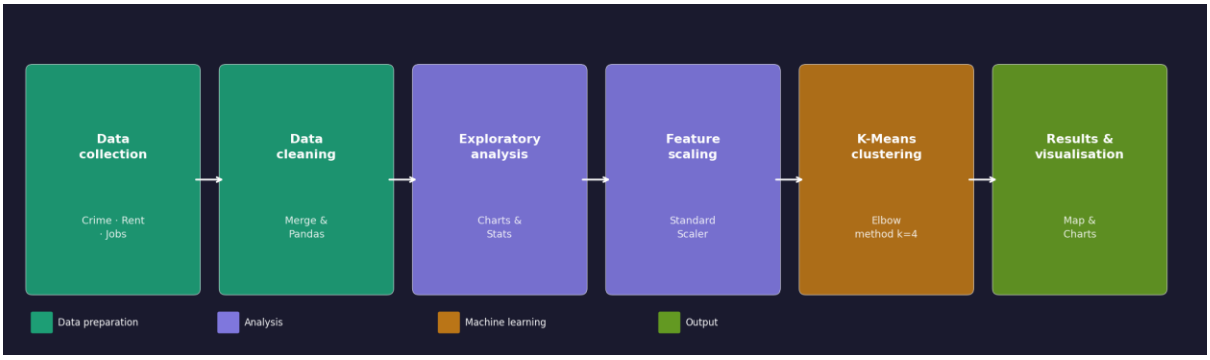

Fig 3.0: Project Flowchart

3.1. Exploratory Data Analysis

3.1.1. Descriptive Statistics for Primary Benchmarks

Initial data analysis was performed on all three datasets. The average unemployment rate across Munich's 25 districts is 4.7%. The average crime rate is 38.4 crimes per 1,000 residents. The most affordable average rent is EUR 1,180/month (Allach-Untermenzing) while the most expensive is EUR 2,100/month (Altstadt-Lehel). That is a difference of EUR 920/month, nearly EUR 11,000 per year.

3.1.2. Key Observations

- Feldmoching-Hasenbergl has the highest unemployment rate at 7.1% - more than double that of Bogenhausen at 2.8%
- Altstadt-Lehel has the highest crime rate at 98.2 per 1,000 residents - driven primarily by city centre activity and tourism
- Rent in Altstadt-Lehel at EUR 2,100/month is 78% more expensive than Allach-Untermenzing at EUR 1,180/month

3.2. Machine Learning Algorithm

A clustering algorithm called K-Means was used to group the districts in order of desirability for new immigrants. K-Means is an unsupervised Machine Learning algorithm that groups data into k number of clusters. This method uses a centroid based algorithm to group districts into k clusters such that all districts with similar characteristics are in the same cluster. The algorithm works in the following steps:

- i) Determine most optimal k (number of clusters)
- ii) Initialize k such that initial means are randomly generated within the data domain
- iii) k clusters are created by associating every observation with the nearest mean
- iv) The centroid of each of the k clusters becomes the new mean
- v) Steps iii and iv are repeated until convergence is reached

3.2.1. Determining Optimum Number of Clusters

The optimal k was obtained using the Elbow Method. The dataset was fit with the K-Means model for a range of values (1-10). The inertia for each value of k was stored and plotted on a line chart. The point where the curve bends is the optimal k. The optimum number of clusters obtained was k = 4, meaning all 25 Munich districts would be grouped into 4 clusters.

3.2.2. Feature Scaling

Before clustering, StandardScaler from scikit-learn was used to normalise all three variables. This is a critical step - without it, rent (ranging from 1,180 to 2,100) would dominate the clustering over unemployment (ranging from 2.8 to 7.1) simply due to the difference in scale, not importance.

3.2.3. Clustering Districts by Primary and Secondary Benchmarks

The districts were grouped using all three indicators. The outcome was used to generate the final Munich District Desirability Index. The cluster results are shown in Table 1.0.

Cluster	Avg Unemployment	Avg Crime Rate	Avg Rent (EUR)	Label
0	5.71%	42.81	1,395	Least Desirable
1	3.63%	34.89	1,864	Desirable
2	4.20%	98.20	2,100	Semi-Desirable
3	4.02%	27.38	1,502	Most Desirable

Table 1.0: Result of Clustering using Primary and Secondary Benchmarks

4. Results and Discussion

Using the steps explained in Section 3, all 25 Munich districts were ranked into four desirability categories. The final Munich District Desirability Index chart is shown in Fig 4.0.

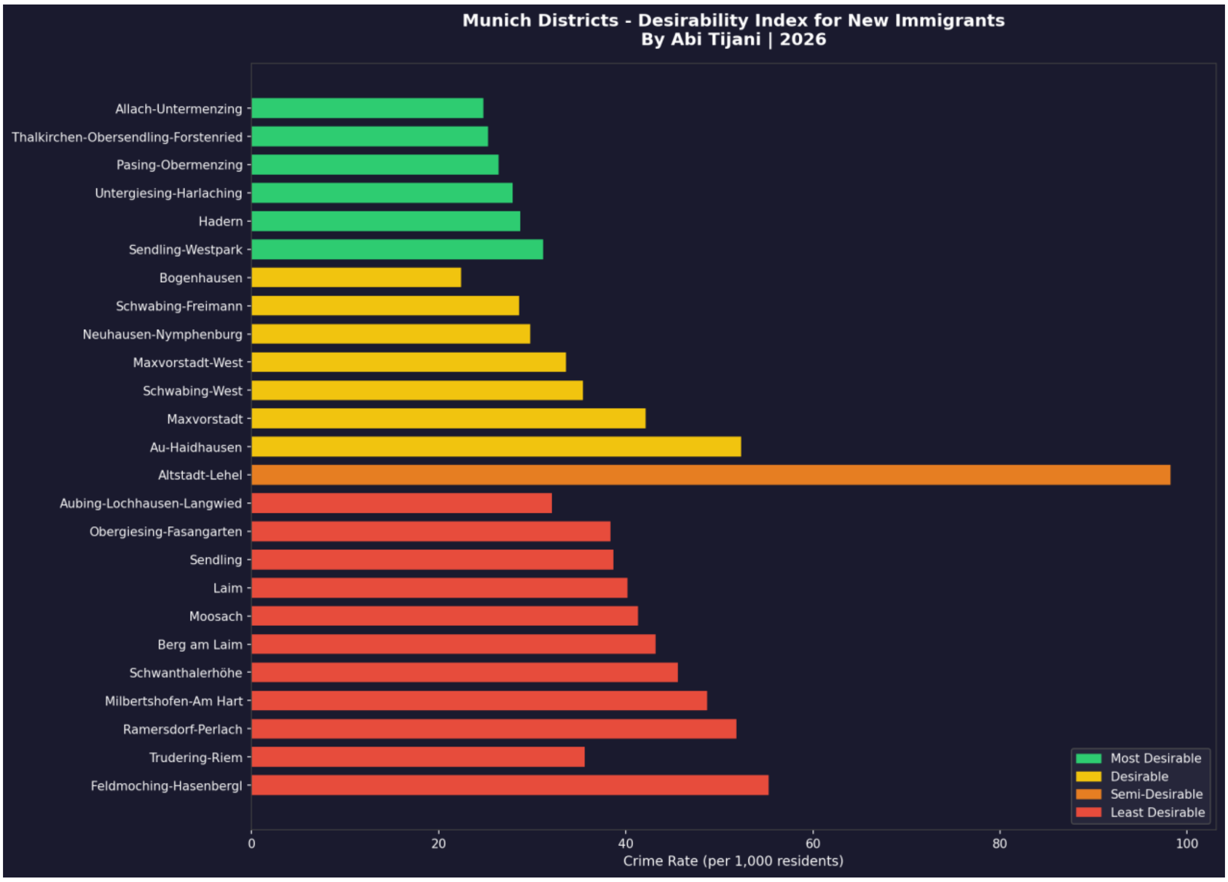

Fig 4.0: Munich Districts Desirability Index

4.1. Most Desirable Districts

Sendling-Westpark, Hadern, Untergiesing-Harlaching, Pasing-Obermenzing, Thalkirchen-Obersendling-Forstenried, Allach-Untermenzing

These 6 districts have the lowest crime rates in Munich averaging just 27.4 crimes per 1,000 residents, moderate unemployment averaging 4.0%, and affordable rents averaging EUR 1,502/month. For a new immigrant prioritising safety and value for money, these districts offer the strongest combination of all three factors.

4.2. Desirable Districts

Bogenhausen, Maxvorstadt, Schwabing-West, Schwabing-Freimann, Neuhausen-Nymphenburg, Au-Haidhausen, Maxvorstadt-West

These 7 districts have the lowest unemployment rates in Munich averaging just 3.6% and vibrant communities. The trade-off is higher average rent at EUR 1,864/month.

4.3. Semi-Desirable Districts

Altstadt-Lehel - Munich's historic city centre with the highest crime rate at 98.2 per 1,000 residents and highest rent at EUR 2,100/month. Crime is largely driven by tourist-area activity rather than resident-driven crime.

4.4. Least Desirable Districts

Feldmoching-Hasenbergl, Ramersdorf-Perlach, Milbertshofen-Am Hart, Berg am Laim, Schwanthalerhöhe, Moosach, Laim, Sendling, Obergiesing-Fasangarten, Trudering-Riem, Aubing-Lochhausen-Langwied

These 11 districts have the highest unemployment averaging 5.7% and highest crime among residential districts averaging 42.8 per 1,000. Rents average EUR 1,395/month.

5. Conclusion

From the results, we can make the following deductions:

- Only 24% of Munich's districts fall into the Most Desirable category, making those 6 districts genuinely worth targeting as a new immigrant
- Most Desirable Neighbourhoods: Consider Sendling-Westpark, Hadern or Pasing-Obermenzing for the best combination of low crime, affordable rent and decent employment rates
- Looking for a Vibrant Community: Bogenhausen and Neuhausen-Nymphenburg offer the best quality of life in the Desirable category with lowest unemployment in Munich
- The City Centre Trap: Altstadt-Lehel is exciting but at EUR 2,100/month with the highest crime rate, new immigrants pay a very high premium for the postcode
- Avoid if You Can: Feldmoching-Hasenbergl shows the weakest combination of all three metrics. The savings on rent do not justify the difference in safety and employment access

References

[1] Destatis - German Federal Statistical Office, Net Migration Report 2024: https://www.destatis.de

[2] Mercer Quality of Living City Ranking 2025: https://www.mercer.com/insights/quality-of-living/

[3] Jessica Ayodele - Analysis of Toronto Neighbourhoods using Machine Learning, Towards Data Science, 2020

[4] Polizeiprasidium Munchen - Kriminalitatsatlas Munchen 2023: https://www.polizei.bayern.de/muenchen

[5] Statistisches Amt Munchen - Munchner Sozialatlas 2023: https://www.muenchen.de/rathaus/stadtinfos/statistik

[6] Immoscout24 - Mietpreise Munchen nach Stadtbezirk 2024: https://www.immoscout24.de

[7] OpenStreetMap contributors: https://www.openstreetmap.org

[8] Google Colab Notebook: https://github.com/AbiXData/munich-neighbourhood-analysis

[9] GitHub Repository: https://github.com/AbiXData/munich-neighbourhood-analysis